

Name: Narendra Kumar S

Date: 23-07-2025

Task:

1. A library needs to develop an online application for two types of users/roles, Adults and children.

Both of these users should be able to register an account.

Any user who is less than 12 years of age will be registered as a child and they can borrow a "Kids" category book

for 10 days, whereas an adult can borrow "Fiction" category books which need to be returned within 7 days.

1.1.Create an interface LibraryUser with the following methods declared,

Method Name

registerAccount

requestBook

1.2.Create 2 classes "KidUser" and "AdultUser" which implements the LibraryUser interface.

1.3.Both the classes should have two instance variables as specified below.

age	int
bookType	String

1.4.The methods in the KidUser class should perform the following logic.

1.4.1. registerAccount : if age < 12, a message displaying "You have successfully registered under a Kids Account" should be displayed in the console.

If(age>12), a message displaying, "Sorry, Age must be less than 12 to register as a kid" should be displayed in the console.

1.4.2. requestBook : if bookType is "Kids", a message displaying "Book Issued successfully, please return the book within 10 days" should be displayed in the console.

else, a message displaying, "You are allowed to take only kids books" should be displayed in the console.

1.5.The methods in the AdultUser class should perform the following logic.

1.5.1. registerAccount : if age > 12, a message displaying "You have successfully registered under an Adult Account" should be displayed in the console.

If age<12, a message displaying, "Sorry, Age must be greater than 12 to register as an adult" should be displayed in the console.

1.5.2. requestBook : if bookType is "Fiction", a message displaying "Book Issued successfully, please return the book within 7 days" should be displayed in the console.

else, a message displaying, "

You are allowed to take only adult Fiction books" should be displayed in the console.

1.6.Create a class LibraryInterfaceDemo with a main method which performs the below functions,

In the main method, test all the methods.

Solution:

```
interface LibraryUser {
    void registerAccount();
    void requestBook();
}
class KidUser implements LibraryUser {

    int age;
    String bookType;
    KidUser(int age, String bookType) {
        this.age = age;
        this.bookType = bookType;
    }
    @Override
    public void registerAccount() {
        if (age < 12) {
            System.out.println("You have successfully registered under a Kids Account");
        } else {
            System.out.println("Sorry, Age must be less than 12 to register as a kid");
        }
    }
    @Override
    public void requestBook() {
        if ("Kids".equalsIgnoreCase(bookType)) {
            System.out.println("Book Issued successfully, please return the book within 10 days");
        } else {
            System.out.println("You are allowed to take only kids books");
        }
    }
    @Override
    public String toString() {
        return String.format("KidUser [age=%s, bookType=%s]", age, bookType);
    }
}
class AdultUser implements LibraryUser {
    int age;
    String bookType;
    AdultUser(int age, String bookType) {
        this.age = age;
        this.bookType = bookType;
    }
    @Override
```

```

public void registerAccount() {
    if (age > 12) {
        System.out.println("You have successfully registered under an Adult Account");
    } else {
        System.out.println("Sorry, Age must be greater than 12 to register as an adult");
    }
}

@Override
public void requestBook() {
    if ("Fiction".equalsIgnoreCase(bookType)) {
        System.out.println("Book Issued successfully, please return the book within 7 days");
    } else {
        System.out.println("You are allowed to take only adult Fiction books");
    }
}

@Override
public String toString() {
    return String.format("AdultUser [age=%s, bookType=%s]", age, bookType);
}
}

public class LibraryInterfaceDemo {
    public static void main(String[] args) {
        KidUser kid1 = new KidUser(10, "Kids");
        System.out.println(kid1.toString());
        kid1.registerAccount();
        kid1.requestBook();

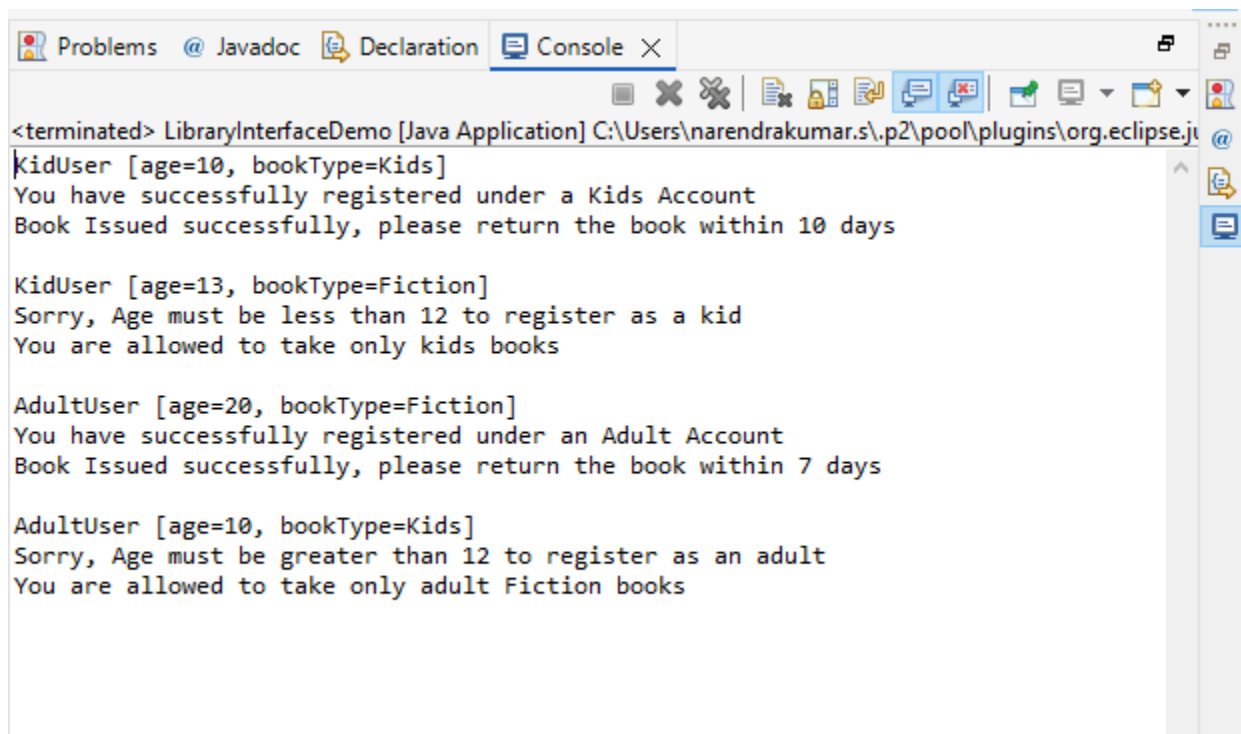
        System.out.println();
        KidUser kid2 = new KidUser(13, "Fiction");
        System.out.println(kid2);
        kid2.registerAccount();
        kid2.requestBook();

        System.out.println();
        AdultUser adult1 = new AdultUser(20, "Fiction");
        System.out.println(adult1);
        adult1.registerAccount();
        adult1.requestBook();

        System.out.println();
        AdultUser adult2 = new AdultUser(10, "Kids");
        System.out.println(adult2);
        adult2.registerAccount();
        adult2.requestBook();
    }
}

```

Output:

The screenshot shows the Eclipse IDE's Console window. The title bar includes tabs for 'Problems', '@ Javadoc', 'Declaration', and 'Console'. The console output is as follows:

```
<terminated> LibraryInterfaceDemo [Java Application] C:\Users\narendrakumar.s\p2\pool\plugins\org.eclipse.jdt
KidUser [age=10, bookType=Kids]
You have successfully registered under a Kids Account
Book Issued successfully, please return the book within 10 days

KidUser [age=13, bookType=Fiction]
Sorry, Age must be less than 12 to register as a kid
You are allowed to take only kids books

AdultUser [age=20, bookType=Fiction]
You have successfully registered under an Adult Account
Book Issued successfully, please return the book within 7 days

AdultUser [age=10, bookType=Kids]
Sorry, Age must be greater than 12 to register as an adult
You are allowed to take only adult Fiction books
```

2. Write a program to read two integer array lists of size 5 each as input and to merge the two arrayLists, sort the merged arraylist in ascending order and fetch the elements at 2nd, 6th and 8th index into a new arrayList and return the final ArrayList.

Solution:

```
import java.util.*;
public class MergeandFetch {
    public static ArrayList<Integer> mergeSortAndFetch(List<Integer> list1, List<Integer> list2) {
        ArrayList<Integer> list3 = new ArrayList<>();
        list3.addAll(list1);
        list3.addAll(list2);
        Collections.sort(list3);
        ArrayList<Integer> result2 = new ArrayList<>();
        int[] indexes = {2, 6, 8};
        for (int num : indexes) {
```

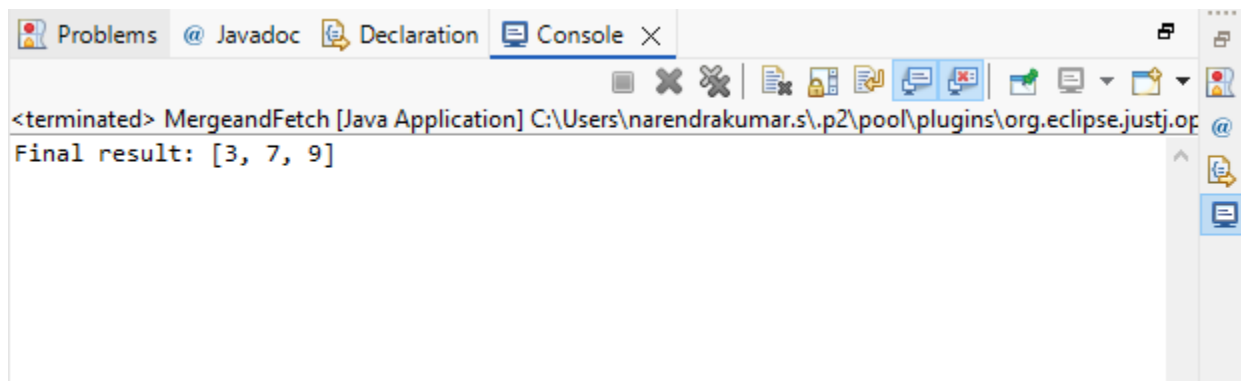
```

        if (num < list3.size()) {
            result2.add(list3.get(num));
        }
    }
    return result2;
}

public static void main(String[] args) {
    List<Integer> list1 = Arrays.asList(10, 5, 3, 7, 8);
    List<Integer> list2 = Arrays.asList(2, 9, 6, 1, 4);
    ArrayList<Integer> result1 = mergeSortAndFetch(list1, list2);
    System.out.println("Final result: " + result1);
}
}

```

Output:



3. Read student details as input. The details would include name, mark in the given order. The datatype for name is string, mark is float. Create a hashmap that contains name as key and mark as value. Get student name as input and display the student grade.

1. If Mark is less than 60, then grade is FAIL.
2. If Mark is greater than or equal to 60, then grade is PASS.

Solution:

```

import java.util.*;

public class StudentGrade {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        HashMap<String, Float> studentMap = new HashMap<>();
        for (int i = 0; i < 3; i++) {
            System.out.print("Enter student name: ");
            String name = sc.nextLine();

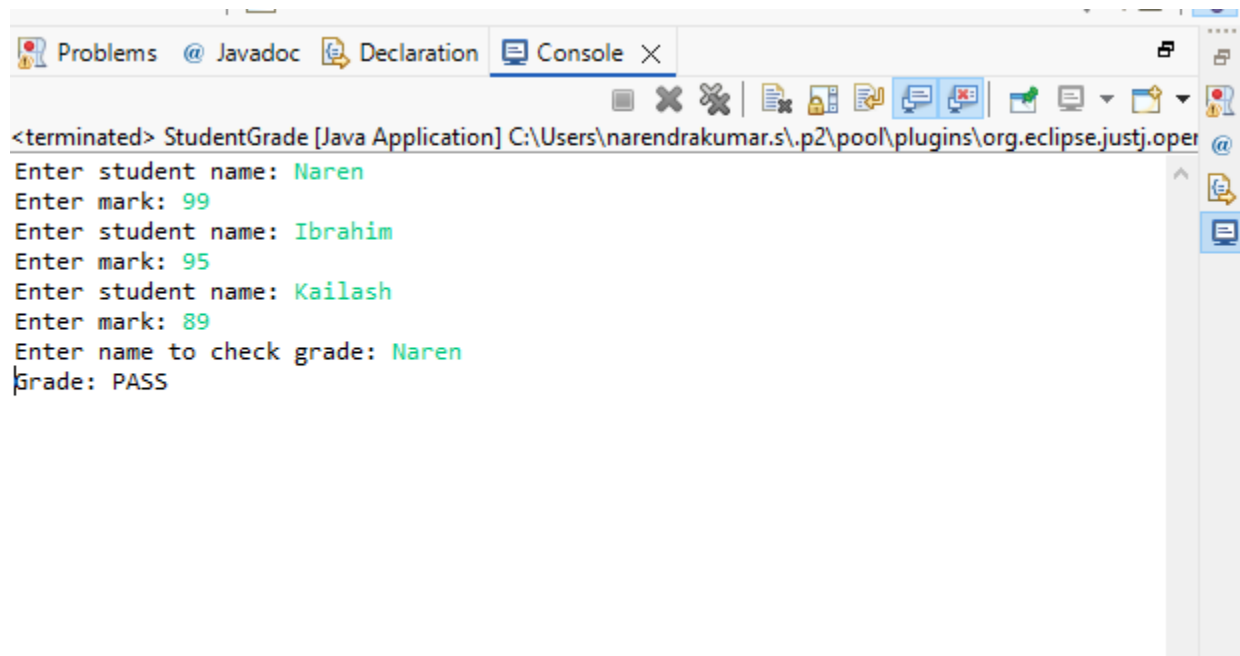
```

```

        System.out.print("Enter mark: ");
        float mark = sc.nextFloat();
        sc.nextLine();
        studentMap.put(name, mark);
    }
    System.out.print("Enter name to check grade: ");
    String searchName = sc.nextLine();
    if (studentMap.containsKey(searchName)) {
        float mark = studentMap.get(searchName);
        if (mark < 60) {
            System.out.println("Grade: FAIL");
        } else {
            System.out.println("Grade: PASS");
        }
    } else {
        System.out.println("Student not found!");
    }
}
}
}

```

Output:



```

<terminated> StudentGrade [Java Application] C:\Users\narendrakumar.s\p2\pool\plugins\org.eclipse.justj.open
Enter student name: Naren
Enter mark: 99
Enter student name: Ibrahim
Enter mark: 95
Enter student name: Kailash
Enter mark: 89
Enter name to check grade: Naren
Grade: PASS

```

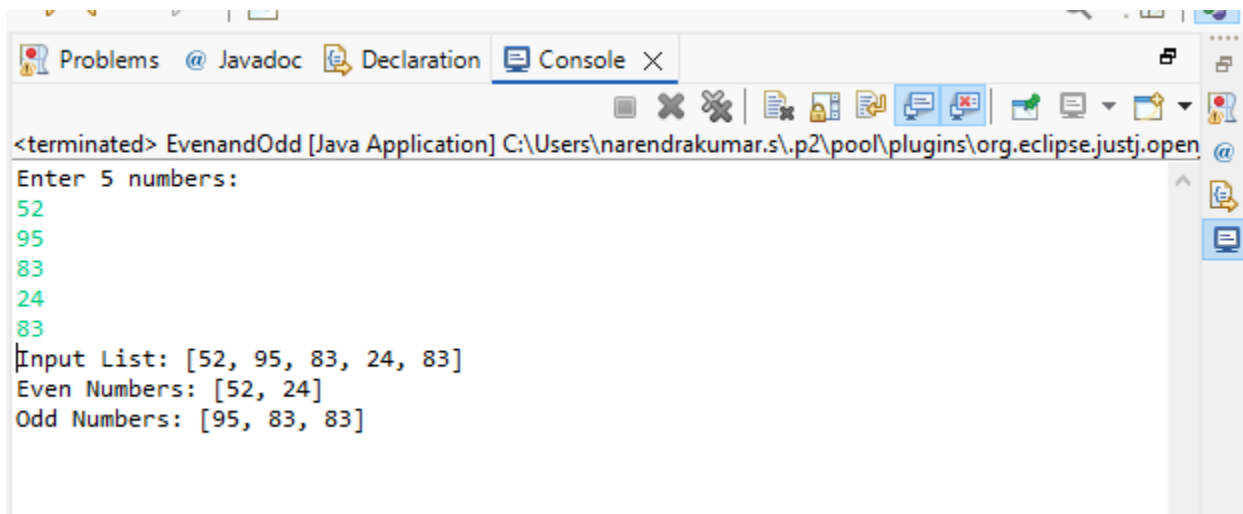
4. Write a program to get integers as input and store in the arraylist. Traverse the input list, if the number is even store in a arraylist called evenNumbersList and oddnumbers in oddNumberList.

Print the input list and the lists containing even numbers and odd numbers.

Solution:

```
import java.util.*;
public class EvenandOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        ArrayList<Integer> inputList = new ArrayList<>();
        ArrayList<Integer> evenList = new ArrayList<>();
        ArrayList<Integer> oddList = new ArrayList<>();
        System.out.println("Enter 5 numbers:");
        for (int i = 0; i < 5; i++) {
            int num = sc.nextInt();
            inputList.add(num);
        }
        for (int num : inputList) {
            if (num % 2 == 0) {
                evenList.add(num);
            } else {
                oddList.add(num);
            }
        }
        System.out.println("Input List: " + inputList);
        System.out.println("Even Numbers: " + evenList);
        System.out.println("Odd Numbers: " + oddList);
    }
}
```

Output:



The screenshot shows the Eclipse IDE's console window. The title bar indicates the application is 'EvenandOdd [Java Application]' running at 'C:\Users\narendrakumar.s\p2\pool\plugins\org.eclipse.justj.open'. The console output is as follows:

```
<terminated> EvenandOdd [Java Application] C:\Users\narendrakumar.s\p2\pool\plugins\org.eclipse.justj.open
Enter 5 numbers:
52
95
83
24
83
Input List: [52, 95, 83, 24, 83]
Even Numbers: [52, 24]
Odd Numbers: [95, 83, 83]
```

